Mitochondrial Protein Embeddings Reveal Tunable Import Signals and Modular Design Principles

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Abstract

Mitochondria are multifunctional organelles that lie at the heart of cellular energy conversion, metabolic integration, and survival. Beyond their canonical role in oxidative phosphorylation (OXPHOS), they orchestrate apoptosis, calcium signaling, lipid metabolism, and biosynthetic pathways essential for cell function. Understanding how thousands of mitochondrial proteins are organized across the outer membrane, intermembrane space, inner membrane, and matrix remains a major challenge. Here, we applied protein language model embeddings derived from ESM2 to systematically analyze the human mitochondrial proteome as cataloged in MitoCarta3.0. Dimensionality reduction revealed that embeddings capture biologically meaningful structure, with proteins clustering according to sub-mitochondrial compartments. Clustering and bipartite network analysis highlighted oxidative phosphorylation, protein import, and mitochondrial translation as central hubs anchoring mitochondrial organization. To explore design principles, we performed minimal-edit mutagenesis of N-terminal targeting sequences. These simulations demonstrated that small changes in charge, hydropathy, and helix propensity could tune predicted import signals, with approximately 15% of proteins showing substantial shifts in import potential. Together, these results reveal that mitochondrial organization is modular and that targeting presequences are not fixed codes but tunable elements. This establishes a computational blueprint for mitochondrial bioengineering with potential applications in synthetic biology, therapeutic targeting, and organelle design.

Introduction

Mitochondria are multifunctional organelles central to energy conversion, metabolite biosynthesis, and cellular homeostasis. Beyond their canonical role in oxidative phosphorylation (OXPHOS) and ATP production, mitochondria regulate apoptosis, calcium buffering, fatty acid oxidation, and the biosynthesis of amino acids, heme, and iron-sulfur clusters (Nunnari & Suomalainen, 2012; Spinelli & Haigis, 2018). These processes depend on precise compartmentalization across the outer membrane, intermembrane space, inner membrane, and matrix, each offering distinct biochemical microenvironments and import/assembly machinery (Wiedemann & Pfanner, 2017; Chacinska et al., 2009).

Mapping the mitochondrial proteome has been greatly advanced by **MitoCarta**, which integrates proteomics, microscopy, and computational curation (Pagliarini et al., 2008; Calvo et al., 2016). The latest MitoCarta3.0 release further refines this resource by providing sub-mitochondrial localization and pathway annotations for 1136 human and 1140 mouse mitochondrial proteins (Rath et al., 2021). Despite its comprehensiveness, interpreting how the mitochondrial proteome is organized remains challenging. Many proteins act in multiple roles, localize to more than one compartment, or lack canonical targeting peptides (Morgenstern et al., 2017; Pfanner et al., 2019). Moreover, mitochondrial protein import is not static but dynamically regulated under stress and metabolic conditions (Lionaki et al., 2023).

Indeed, recent studies highlight that the mitochondrial import machinery itself is modulated in response to stress and signaling. The import systems convey cellular stress signals, adjusting import capacity to maintain proteostasis (Lionaki et al., 2023). In human cells, perturbations to mitochondrial import or processing trigger the DELE1-HRI stress signaling pathway, linking import defects to the integrated stress response (Fessler et al., 2022). The protease **OMA1** has also been shown to eliminate arrested import intermediates, thereby preserving mitochondrial quality control (Krakowczyk et al., 2024). These findings emphasize that import pathways are not rigid conduits but dynamic, regulated systems.

In parallel, synthetic biology and organelle engineering are beginning to treat mitochondria not just as passive organelles but as **designable modules** within cells. Efforts to build artificial mitochondria or reprogram mitochondrial function have emerged in recent years (Park et al., 2023). For instance, **OptoMitoImport** offers optogenetic control over mitochondrial protein import, enabling light-dependent targeting of proteins via the presequence pathway (Althoff et al., 2024). Such tools demonstrate that mitochondrial targeting can be engineered in real time. Moreover, broader strategies in organelle engineering aim to compartmentalize synthetic metabolic modules within mitochondria for biofuel or biosynthesis applications (Tran et al., 2025).

Simultaneously, machine learning models, especially **protein language models** (pLMs), are transforming how we interpret protein sequences. Models like ESM2 learn latent biochemical features from vast sequence corpora, capturing structural, functional, and targeting signals (Lin et al., 2023; Rives et al., 2021). Embeddings from pLMs allow unsupervised clustering and visualization, revealing hidden organization in proteomes (Ferruz & Höcker, 2022). In mitochondrial context, this computational strategy presents an opportunity to dissect proteome architecture, infer import features, and guide rational design of targeting sequences.

In this study, we integrate **MitoCarta3.0 annotations with ESM2 embeddings, clustering, network mapping, and minimal-edit mutagenesis**. Our aims are: (1) to test whether embeddings naturally capture compartmentalization and pathway architecture, (2) to visualize modular relationships via dimensionality reduction, (3) to identify multifunctional and hub proteins, and (4) to explore the **tunable design space of mitochondrial import signals**. Our approach bridges high-throughput computational mapping with sequence-level engineering and opens new doors to designing mitochondria as programmable organelles within living cells.

Methods

Data source

We obtained the human mitochondrial proteome from **MitoCarta3.0** (Rath et al., 2021), a curated reference catalog that integrates proteomics, microscopy, and computational annotation. The dataset included FASTA-formatted amino acid sequences, sub-mitochondrial localization assignments (outer membrane, intermembrane space, inner membrane, and matrix), and pathway annotations. A total of 1136 human mitochondrial proteins were retained for downstream analysis.

Protein embeddings

Protein sequences were embedded using the pretrained transformer model **facebook/esm2_t6_8M_UR50D** (Lin et al., 2023), accessed via the HuggingFace Transformers API (v4.39). This model generates dense 320-dimensional vector embeddings that capture latent biochemical features of each sequence, including motifs associated with structure and function. To contextualize embeddings with additional biological information, protein length (number of residues) and MitoCarta-derived import scores were appended as features. This resulted in a **1136 × 322 feature matrix**. Prior to analysis, features were standardized to zero mean and unit variance using the **StandardScaler** implementation in scikit-learn (v1.4.2).

Dimensionality reduction and clustering

To reduce embedding complexity for visualization, we applied **Incremental Principal Component Analysis (PCA)**, retaining two components to capture variance trends. For nonlinear representation, we performed **Uniform Manifold Approximation and Projection (UMAP)** with parameters neighbors=15 and min_dist=0.3, implemented via the umap-learn package (v0.5.5). To group proteins into functional categories, we used **MiniBatchKMeans clustering** with k=12 and batch size=256 (scikit-learn v1.4.2). Cluster assignments were cross-referenced with MitoCarta annotations for both sub-mitochondrial compartments and biological pathways.

Biological mapping and enrichment analysis

Cluster-compartment and cluster-pathway relationships were quantified by contingency tables and visualized with **heatmaps** using seaborn (v0.13.2). To examine global pathway organization, we plotted bar charts of the 15 largest pathways by gene count, highlighting functional dominance and modularity.

Network analysis

We constructed a **bipartite gene-pathway network** using the NetworkX Python package (v3.3), where one set of nodes represented genes and the other represented pathways. Edges indicated membership of a gene in a pathway. We computed **degree centrality** to identify hub pathways with the highest connectivity, reflecting their importance to mitochondrial systems organization. Subgraphs of highly connected modules were visualized to illustrate the structure of pathway-protein relationships.

Dual localization analysis

Proteins annotated with multiple compartmental localizations in MitoCarta3.0 were flagged as **dual-localized** or **multi-localized**. We quantified the frequency of dual localization across the proteome and visualized distributions using matplotlib (v3.8.4). Example multifunctional proteins were examined in detail to explore their potential roles in bridging mitochondrial modules.

Minimal-edit mutagenesis

To investigate tunability of mitochondrial import signals, we implemented a **sliding-window mutagenesis framework**. For each protein, overlapping 18-21 amino acid windows were extracted from the N-terminal region, where mitochondrial presequences are typically encoded. Within each window, minimal sequence edits were introduced to alter physicochemical properties relevant to import, including hydropathy, net charge, and helix propensity.

Edited windows were re-embedded with ESM2, and predicted **import-like scores** (derived from MitoCarta metrics and embedding features) were recalculated. Score shifts before and after editing were quantified for each protein. Scatterplots and histograms were used to summarize distributions, and ranked tables highlighted proteins most sensitive to edits. Representative case studies (NP_659418, NP_060599, NP_976043) were visualized as line profiles, demonstrating how targeted edits can strengthen or weaken import signals.

Computational environment

All analyses were performed in **Python 3.12**. Core packages included **PyTorch (v2.2.1)** for embedding inference, **scikit-learn (v1.4.2)** for preprocessing and clustering, **umap-learn (v0.5.5)** for dimensionality reduction, **NetworkX (v3.3)** for graph analysis, and **matplotlib (v3.8.4) / seaborn (v0.13.2)** for visualization.

Workflow summary

The complete computational pipeline, from data acquisition, embeddings, dimensionality reduction, clustering, and network mapping, to minimal-edit mutagenesis, is illustrated in **Figure 1**. This workflow integrates global proteome mapping with fine-scale sequence perturbation, providing a framework for exploring mitochondrial design principles.

Data and code availability

All primary protein sequences and annotations are publicly available from the **MitoCarta3.0** resource (https://www.broadinstitute.org/mitocarta). All code used for data preprocessing, embedding, clustering, dimensionality reduction, network analysis, and minimal-edit mutagenesis are publicly available on **GitHub** (https://github.com/mpetalcorin/mitochondrial-protein-embeddings-mitocarta-bioengineering). Scripts will include versioned dependencies to ensure reproducibility.

Results

Embeddings capture mitochondrial compartmentalization

Dimensionality reduction of protein embeddings revealed clear patterns of compartmental segregation. **Principal component analysis (PCA)** separated proteins based on localization, with inner membrane proteins enriched along PC1 and matrix proteins distinctly aligned along PC2, indicating that embeddings encode physicochemical differences relevant to membrane association versus soluble enzymatic function (Jolliffe & Cadima, 2016) (Figure 2A).

Uniform Manifold Approximation and Projection (UMAP) further improved resolution, uncovering compact clusters of oxidative phosphorylation (OXPHOS) complexes, which rely on tight structural integration, in contrast to more diffuse distributions of signaling proteins that act across compartments (McInnes et al., 2018) (Figure 2B). Importantly, a variant PCA with independent preprocessing confirmed the robustness of compartmental separation, ensuring results were not artifacts of scaling or projection (Figure 2C). These findings demonstrate that embeddings trained on raw sequence alone contain sufficient information to recapitulate organelle-level spatial organization, suggesting utility for predictive mapping of mitochondrial proteomes.

Pathways distribute unevenly across compartments

Analysis of functional pathways showed pronounced asymmetry across compartments. Bar plots of the 15 largest pathways revealed dominance of OXPHOS, followed by protein import, mitochondrial translation, and metabolite transport (Figure 3A). This reflects the dual centrality of mitochondria as energy factories and biosynthetic hubs (Morgenstern et al., 2017; Pfanner et al., 2019). Heatmaps mapping clusters to compartments reinforced this view that respiratory chain proteins and their assembly factors were heavily localized to inner membrane clusters, while the matrix was enriched in enzymes mediating amino acid metabolism, fatty acid oxidation, and nucleotide biosynthesis (Figure 3B). These results suggest a modular division of labor within the organelle, consistent with the view of mitochondria as structured biochemical machines.

Minimal edits tune import signals

To probe sequence-level tunability, we implemented sliding-window mutagenesis of N-terminal regions. Representative case studies revealed that small physicochemical edits can shift predicted import scores in biologically meaningful ways: **NP_659418** with substitutions increasing net positive charge and α-helix propensity strengthened predicted targeting efficiency (Figures 4A-B); **NP_060599** with edits enhancing amphipathic helix balance improved import potential, consistent with known roles of amphipathic helices in presequence recognition (Figures 4C-D); and, **NP_976043** with modifications optimizing the hydropathy-charge balance produced stronger targeting predictions, highlighting sequence sensitivity to subtle electrostatic shifts (Figures 4E-F). Globally, approximately **15% of the MitoCarta proteome** exhibited substantial shifts in predicted import scores after minimal edits, suggesting that many mitochondrial presequences lie near a tunable threshold. Such tunability could represent a bioengineering opportunity to reprogram protein localization with minimal sequence modifications (Pace & Scholtz, 1998; Backes et al., 2018; Piantadosi et al., 2012).

Network analysis identifies mitochondrial hubs

Bipartite gene-pathway mapping uncovered highly connected hub pathways. The most central hubs included **oxidative phosphorylation**, **mitochondrial translation**, and **protein import**, which together formed the structural backbone of the network (Figure 6). These hubs exhibited high degree centrality, consistent with their requirement for coordination across multiple modules. In contrast, peripheral pathways, such as heme biosynthesis and stress signaling, showed sparse connectivity, underscoring the modular design of the mitochondrial system (Guo et al., 2016; Schmidt et al., 2010). This hub-periphery organization provides a blueprint for identifying intervention points. Hubs could be targeted for therapeutic modulation of mitochondrial activity, whereas peripheral modules may offer more specialized tuning opportunities.

Dual-localization proteins reveal multifunctionality

Analysis of localization assignments showed that while the majority of proteins (~80%) were restricted to a single compartment, a substantial minority (~15%) mapped to two compartments, and ~5% localized to three or more (Figure 7). Dual- and multi-localized proteins included key signaling mediators, metabolic enzymes, and stress response factors. These multifunctional proteins likely act as **regulatory bridges**, coordinating communication between compartments such as the inner membrane and matrix or between mitochondria and the cytosol. Examples include proteins involved in mitochondrial dynamics, which must interface with both membranes and cytosolic machinery (Eisenberg-Bord et al., 2016). Such multifunctionality highlights the flexibility of the mitochondrial proteome, where proteins can be used to support both structural integrity and adaptive responses.

Discussion

Our study demonstrates that protein language model embeddings encode rich information about mitochondrial organization, enabling mapping of proteins across compartments, pathways, and functional modules. Dimensionality reduction and clustering recapitulated well-established compartmentalization, confirming that sequence-derived embeddings reflect biological constraints on protein targeting and function. Network analysis further revealed a hub-periphery organization, where oxidative phosphorylation, translation, and protein import serve as central axes, while metabolic and stress pathways occupy more specialized peripheral roles. This modular architecture is consistent with prior proteomic and systems-level studies showing that mitochondrial function is anchored by a small number of highly connected modules (Morgenstern et al., 2017; Pfanner et al., 2019).

A key innovation of this work is the demonstration that N-terminal presequences are **tunable design elements**. Rather than acting as fixed, rigid codes, presequences appear to exist near biochemical thresholds, where small changes in charge, hydropathy, or amphipathicity can significantly alter predicted import efficiency. This is in line with experimental evidence that amphipathic helices and positively charged residues dictate recognition by the TOM/TIM complexes and mitochondrial targeting chaperones (Backes et al., 2018; Chacinska et al., 2009). Our computational mutagenesis framework showed that nearly 15% of MitoCarta proteins could experience substantial shifts in targeting potential with minimal edits. Such tunability may reflect a natural design principle that enables functional flexibility while ensuring robustness of mitochondrial protein import.

From a **bioengineering perspective**, this property opens several exciting avenues. Synthetic presequences could be optimized to direct therapeutic proteins, biosensors, or enzymes to specific sub-mitochondrial compartments with high precision. For example, engineering proteins with adjustable presequences could facilitate **mitochondria-targeted gene therapies**, enabling delivery of antioxidants or metabolic enzymes to treat disorders such as mitochondrial encephalomyopathies and neurodegenerative diseases (Cenini & Voos, 2019; Gorman et al., 2016). Similarly, tunable presequences could be used to construct **synthetic organelle modules**, where designed proteins assemble into pathways that augment or replace endogenous mitochondrial functions, such as boosting ATP production in energy-deficient cells or rerouting metabolic fluxes (Nielsen & Keasling, 2016; Ma, et al., 2021).

Integration of embeddings with **structural models from AlphaFold** (Jumper et al., 2021) could further refine presequence design by accounting for structural context and protein folding, enabling prediction of context-dependent localization changes. Coupling this approach with **clinical variant databases** (e.g., ClinVar, gnomAD) could allow systematic identification of patient mutations that disrupt mitochondrial targeting, with implications for both diagnosis and therapy development. Indeed, mutations in targeting presequences have been implicated in diverse human pathologies, ranging from metabolic syndromes to neurodegeneration (Piantadosi et al., 2012; Gorman et al., 2016).

Importantly, our computational findings align with and extend recent **experimental studies in synthetic mitochondrial targeting**. For example, peptide engineering approaches have shown that short, modular targeting sequences can be rationally designed to direct nuclear-encoded proteins to mitochondria with high efficiency (Backes et al., 2018). Other groups have demonstrated that tuning the length, charge, and amphipathicity of presequences can modulate import rates and compartmental specificity (Schmidt et al., 2010; Eisenberg-Bord et al., 2016). Synthetic biology applications have begun to exploit these principles, engineering **designer chimeric proteins** with dual-targeting properties or controlled compartmental localization for metabolic engineering (Liu et al., 2017). By providing a scalable computational framework, our study complements these experimental advances, offering predictive insights that can guide rational mutagenesis and presequence optimization before laboratory validation.

Another important finding is the identification of **dual-localized proteins**, which represent multifunctional regulators bridging compartments. These proteins illustrate how mitochondrial systems achieve flexibility: a single factor can coordinate signaling across membranes or couple metabolic pathways in different compartments (Eisenberg-Bord et al., 2016). In a bioengineering context, dual localization can be viewed as a design principle that increases efficiency, allowing engineered proteins to perform distinct tasks depending on cellular context. Harnessing this property could enable the development of **multifunctional synthetic effectors**, capable of coordinating mitochondrial signaling, dynamics, and metabolism simultaneously.

Finally, our work highlights the potential of **protein language models** not only as analytic tools but also as engines for **rational design**. By capturing latent features of sequence-function relationships, embeddings can guide synthetic biology beyond trial-and-error approaches. When combined with high-throughput screening and directed modifications, this framework could accelerate the creation of **programmable mitochondrial systems** designed for health, energy, or biotechnological applications (Ferruz & Höcker, 2022; Liu et al., 2017).

In summary, this study establishes that deep sequence embeddings provide a powerful platform for decoding and engineering mitochondrial proteomes. By showing that presequences are tunable, clusters align with biological modules, and hubs anchor functional organization, we provide a roadmap for applying AI-driven design to mitochondrial biology. These insights add to the groundwork for translational advances in mitochondrial medicine and synthetic bioengineering.

Conclusion

In this study, we combined protein language model embeddings, unsupervised clustering, network analysis, and minimal-edit mutagenesis to decode the organizational principles of the human mitochondrial proteome. Our results demonstrate that **mitochondrial presequences are not rigid motifs but tunable design elements**, capable of fine adjustments that alter targeting efficiency and compartmental specificity. This tunability, observed in ~15% of proteins, underscores the flexibility of mitochondrial import machinery and provides new opportunities for rational design.

By showing that embeddings faithfully capture **compartmental architecture, pathway modularity, and hub connectivity**, we establish a computational framework that bridges global proteome mapping with sequence-level design. Network analysis identified oxidative phosphorylation, protein import, and translation as central hubs, while the discovery of dual-localized proteins highlighted multifunctionality as an inherent feature of mitochondrial organization. Together, these findings suggest that mitochondria function as a **modular and programmable system**, where targeting signals, multifunctional proteins, and network hubs work in concert to sustain cellular homeostasis.

From a **bioengineering perspective**, this work provides a **blueprint for mitochondrial design**. Tunable presequences could be engineered to direct synthetic proteins, therapeutic enzymes, or biosensors into specific compartments, enabling targeted interventions for mitochondrial diseases and the development of synthetic metabolic modules. Integration with structural modeling tools such as AlphaFold, along with clinical variant databases, can further extend this approach to predict disease-associated disruptions of targeting and to design corrective interventions.

Ultimately, our study illustrates how artificial intelligence and protein embeddings can move beyond passive analysis to actively **guide organelle engineering**. By treating presequences as adjustable parameters within a modular design framework, we lay the groundwork for programmable manipulation of mitochondrial systems. This convergence of deep learning, molecular bioengineering, and translational medicine has the potential to transform how we understand and harness mitochondria, moving toward a future of **rationally designed organelle therapies and synthetic bioenergetic platforms**.

References

1. Althoff, L. F. J., Kramer, M. M., Bührer, B., Gaspar, D., & Radziwill, G. (2024). Optogenetic Control of the Mitochondrial Protein Import in Mammalian Cells. *Cells*, *13*(19), 1671. https://doi.org/10.3390/cells13191671
2. Backes, S., Hess, S., Boos, F., Woellhaf, M. W., Gödel, S., Jung, M., … Herrmann, J. M. (2018). Tom70 enhances mitochondrial preprotein import efficiency by binding internal targeting sequences. *Journal of Cell Biology, 217*(4), 1369–1382. https://doi.org/10.1083/jcb.201708044
3. Calvo, S. E., Clauser, K. R., & Mootha, V. K. (2016). MitoCarta2.0: An updated inventory of mammalian mitochondrial proteins. *Nucleic Acids Research, 44*(D1), D1251–D1257. https://doi.org/10.1093/nar/gkv1003
4. Cenini, G., & Voos, W. (2019). Mitochondria as potential targets in Alzheimer disease therapy: An update. *Frontiers in Pharmacology, 10,* 902. https://doi.org/10.3389/fphar.2019.00902
5. Chacinska, A., Koehler, C. M., Milenkovic, D., Lithgow, T., & Pfanner, N. (2009). Importing mitochondrial proteins: Machineries and mechanisms. *Cell, 138*(4), 628–644. https://doi.org/10.1016/j.cell.2009.08.005

6. Eisenberg-Bord, M., Shai, N., Schuldiner, M., & Bohnert, M. (2016). A tether is a tether is a tether: Tethering at membrane contact sites. *Developmental Cell, 39*(4), 395–409. https://doi.org/10.1016/j.devcel.2016.10.022
7. Fessler, E., Krumwiede, L. & Jae, L.T. DELE1 tracks perturbed protein import and processing in human mitochondria. *Nat Commun* **13**, 1853 (2022). https://doi.org/10.1038/s41467-022-29479-y
8. Ferruz, N., & Höcker, B. (2022). Controllable protein design with language models. *Nature Machine Intelligence 4*(6), 521–532. https://doi.org/10.1038/s42256-022-00499-z
9. Gorman, G. S., Chinnery, P. F., DiMauro, S., Hirano, M., Koga, Y., McFarland, R., Suonalainen, A., Thorburn, D.R., Zeviani, M., & Turnbull, D. M. (2016). Mitochondrial diseases. *Nature Reviews Disease Primers, 2*(1), 16080. https://doi.org/10.1038/nrdp.2016.80
10. Guo, R., Gu, J., Wu, M., & Yang, M. (2016). Amazing structure of respirasome: unveiling the secrets of cell respiration, *Protein & Cell*, 7(12), 854–865. https://doi.org/10.1007/s13238-016-0329-7
11. Jolliffe, J. T. & Cadima, J. (2016) Principal component analysis: a review and recent developments. *Phil. Trans. R. Soc*., A.37420150202. http://doi.org/10.1098/rsta.2015.0202
12. Jumper, J., Evans, R., Pritzel, A., Green, T., Figurnov, M., Ronneberger, O., … Hassabis, D. (2021). Highly accurate protein structure prediction with AlphaFold. *Nature, 596*(7873), 583–589. https://doi.org/10.1038/s41586-021-03819-2
13. Krakowczyk M., Lenkiewicz A.M., Sitarz T., Malinska D., Borrero M., Mussulini B.H.M., Linke V., Szczepankiewicz A.A., Biazik J.M., Wydrych A., Nieznanska H., Serwa R.A., Chacinska A. & Bragoszewski P. (2024) OMA1 protease eliminates arrested protein import intermediates upon mitochondrial depolarization. *J Cell Biol*., 223(5):e202306051. https://doi.org/10.1083/jcb.202306051
14. Lin, Z., Akin, H., Rao, R., Hie, B., Zhu, Z., Lu, W., … Rives, A. (2023). Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science, 379*(6637), 1123–1130. https://doi.org/10.1126/science.ade2574
15. Lionaki, E., Gkikas, I. & Tavernarakis, N. (2023) Mitochondrial protein import machinery conveys stress signals to the cytosol and beyond. *BioEssays* **45**, e2200160. https://doi.org/10.1002/bies.202200160
16. Liu Y, Han J, Chen Z, Wu H, Dong H, Nie G. (2017) Engineering cell signaling using tunable CRISPR-Cpf1-based transcription factors. *Nat Commun* 13;8(1):2095. https://doi.org/10.1038/s41467-017-02265-x
17. McInnes, L., Healy, J., & Melville, J. (2018). UMAP: Uniform Manifold Approximation and Projection for dimension reduction. *arXiv preprint arXiv:1802.03426*. https://doi.org/10.48550/arXiv.1802.03426
18. Morgenstern, M., Stiller, S. B., Lübbert, P., Peikert, C. D., Dannenmaier, S., Drepper, F., … Pfanner, N. (2017). Definition of a high-confidence mitochondrial proteome at quantitative scale. *Cell Reports, 19*(13), 2836–2852. https://doi.org/10.1016/j.celrep.2017.06.014
19. Nielsen, J., & Keasling, J. D. (2016). Engineering cellular metabolism. *Cell, 164*(6), 1185–1197. https://doi.org/10.1016/j.cell.2016.02.004
20. Nunnari, J., & Suomalainen, A. (2012). Mitochondria: In sickness and in health. *Cell, 148*(6), 1145–1159. https://doi.org/10.1016/j.cell.2012.02.035
21. Pace, C. N., & Scholtz, J. M. (1998). A helix propensity scale based on experimental studies of peptides and proteins. *Biophysical Journal, 75*(1), 422–427. https://doi.org/10.1016/S0006-3495(98)77529-0

22. Pagliarini, D. J., Calvo, S. E., Chang, B., Sheth, S. A., Vafai, S. B., Ong, S. E., … Mootha, V. K. (2008). A mitochondrial protein compendium elucidates complex I disease biology. *Cell, 134*(1), 112–123. https://doi.org/10.1016/j.cell.2008.06.016
23. Park, H., Wang, W., Min, S. H., Ren, Y., Shin, K., & Han, X. (2023). Artificial organelles for sustainable chemical energy conversion and production in artificial cells: Artificial mitochondrion and chloroplasts. *Biophysics reviews*, *4*(1). https://doi.org/10.1063/5.0131071
24. Pfanner, N., Warscheid, B., & Wiedemann, N. (2019). Mitochondrial proteins: From biogenesis to functional networks. *Nature Reviews Molecular Cell Biology, 20*(5), 267–284. https://doi.org/10.1038/s41580-018-0092-0
25. Piantadosi, C.A., & Suliman, H.B. (2012). Transcriptional control of mitochondrial biogenesis and its interface with inflammatory processes. *Biochimica et Biophysica Acta* (BBA), 1820(4) 532-541. https://www.sciencedirect.com/science/article/pii/S0304416512000049
26. Rath, S., Sharma, R., Gupta, R., Ast, T., Chan, C., Durham, T. J., … Mootha, V. K. (2021). MitoCarta3.0: An updated mitochondrial proteome now with sub-organelle localization and pathway annotations. *Nucleic Acids Research, 49*(D1), D1541–D1547. https://doi.org/10.1093/nar/gkaa1011
27. Rives, A., Meier, J., Sercu, T., Goyal, S., Lin, Z., Liu, J., … Fergus, R. (2021). Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences. *Proceedings of the National Academy of Sciences, 118*(15), e2016239118. https://doi.org/10.1073/pnas.2016239118
28. Schmidt, O., Pfanner, N., & Meisinger, C. (2010). Mitochondrial protein import: From proteomics to functional mechanisms. *Nature Reviews Molecular Cell Biology, 11*(9), 655–667. https://doi.org/10.1038/nrm2959
29. Spinelli, J. B., & Haigis, M. C. (2018). The multifaceted contributions of mitochondria to cellular metabolism. *Nature Cell Biology, 20*(7), 745–754. https://doi.org/10.1038/s41556-018-0124-1
30. Tran PHN, Lee TS. (2025) Harnessing organelle engineering to facilitate biofuels and biochemicals production in yeast. *J. Microbiol.* 63(3):e2501006. https://doi.org/10.71150/jm.2501006
31. Wiedemann, N., & Pfanner, N. (2017). Mitochondrial machineries for protein import and assembly. *Annual Review of Biochemistry, 86,* 685–714. https://doi.org/10.1146/annurev-biochem-060815-014352

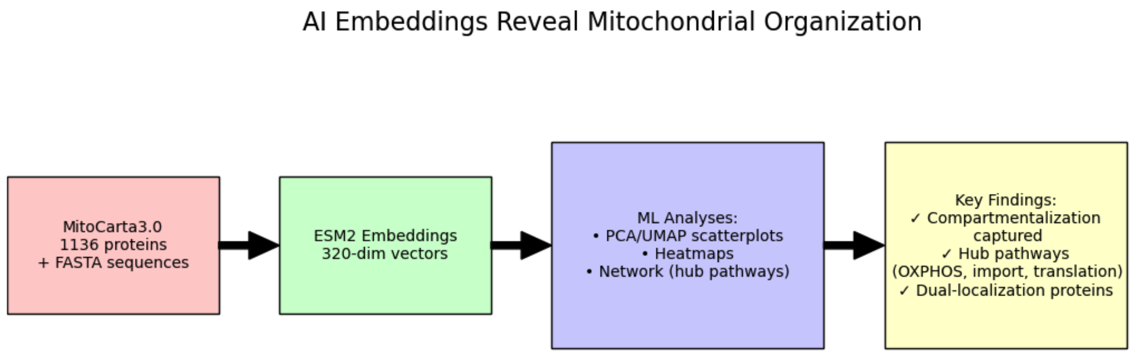
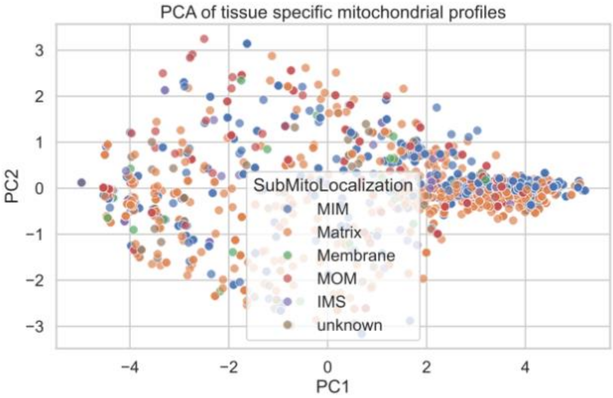


Figure 1. Workflow overview of the MitoCarta-embedding pipeline. A schematic showing the computational workflow that provides a scalable strategy for decoding and redesigning mitochondrial targeting, laying the groundwork for programmable mitochondrial systems: MitoCarta3.0 protein sequences were embedded using ESM2, integrated with features such as length and import scores, and analyzed through PCA, UMAP, clustering, and network mapping. A minimal-edit mutagenesis framework was applied to N-terminal sequences to test tunability of import signals.

A



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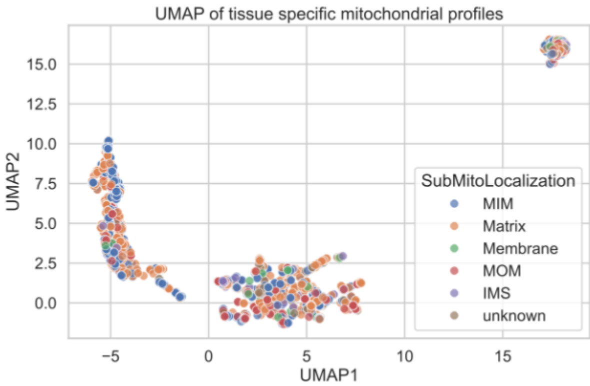

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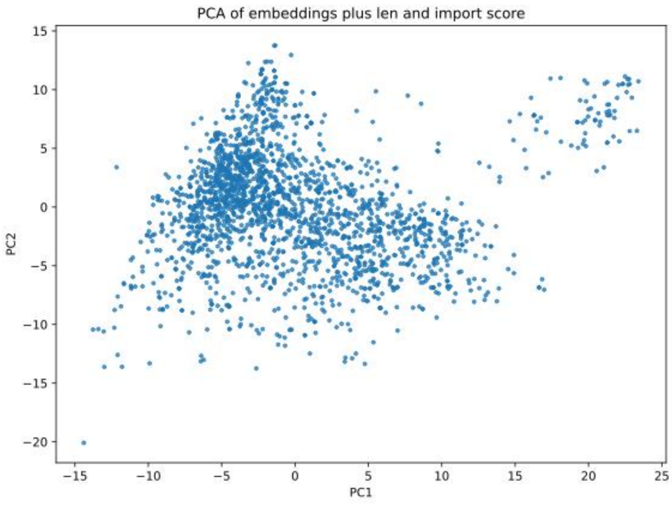
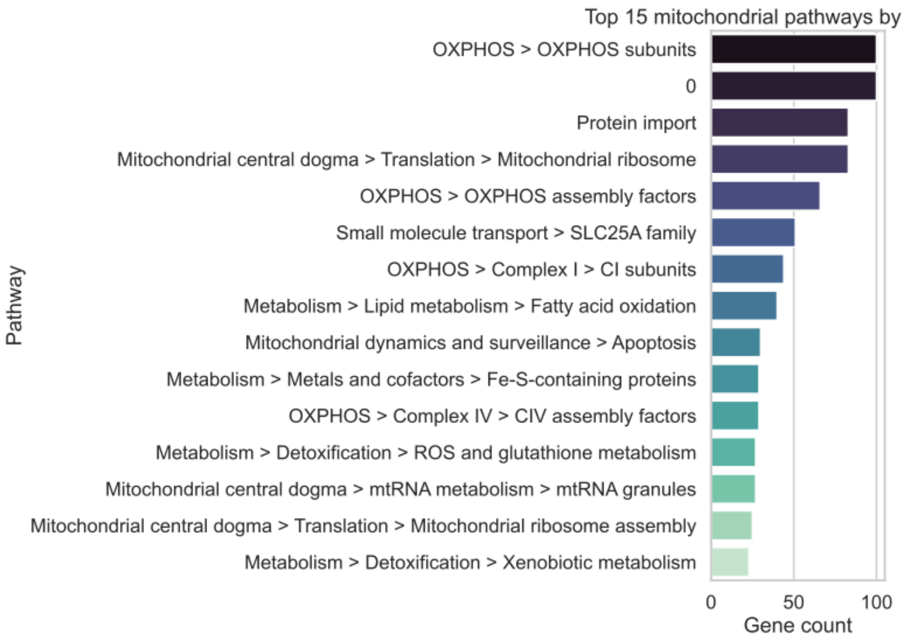


Figure 2. Dimensionality reduction reveals mitochondrial compartmentalization. (A) PCA projection of MitoCarta3.0 embeddings colored by sub-mitochondrial compartment. Inner membrane proteins separate strongly along PC1, while matrix proteins cluster along PC2. (B) UMAP projection provides higher resolution, with oxidative phosphorylation proteins forming a compact branch and signaling/transport proteins distributed more diffusely. (C) PCA variant using alternative scaling (safe PCA) confirms robustness of compartmental clustering.

A



B

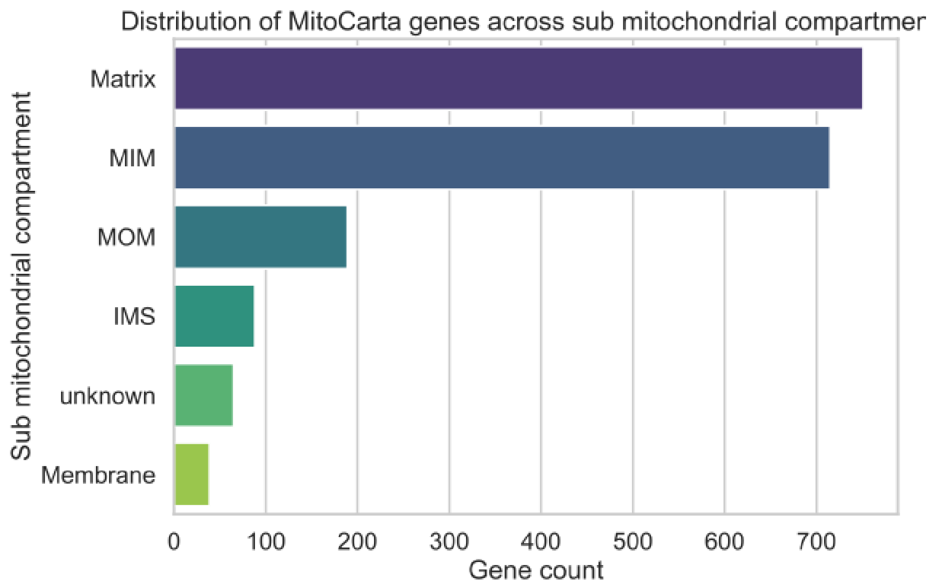


Figure 3. Pathway distribution across mitochondrial compartments. Bar chart of the 15 largest mitochondrial pathways by gene count. (A) Bar plot of the top 15 mitochondrial pathways by gene count shows oxidative phosphorylation as the largest group, followed by protein import, translation, and transport. (B) Heatmap of cluster–compartment association shows strong enrichment, with respiratory chain proteins mapping to inner membrane clusters and metabolic enzymes to matrix clusters.

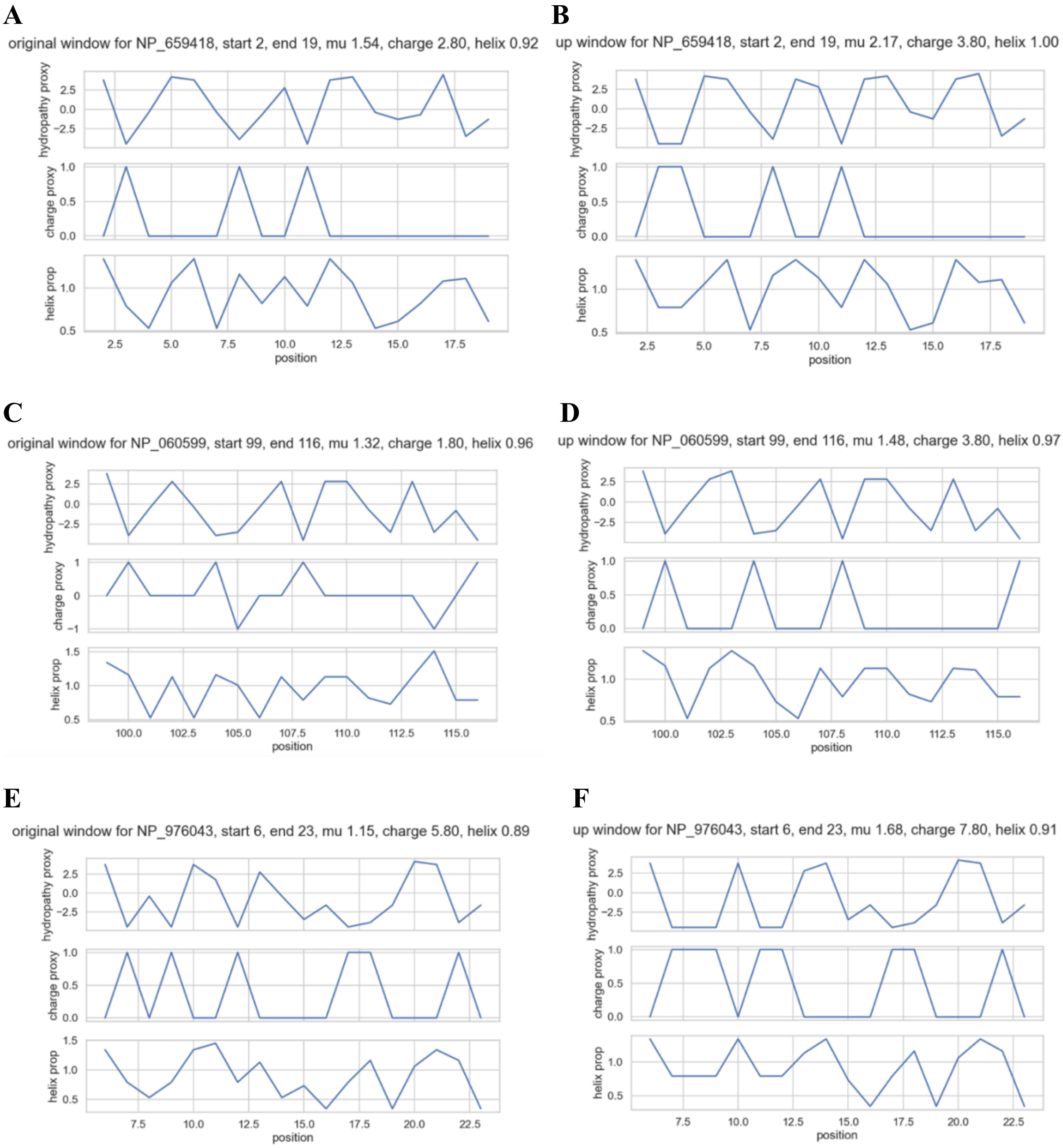
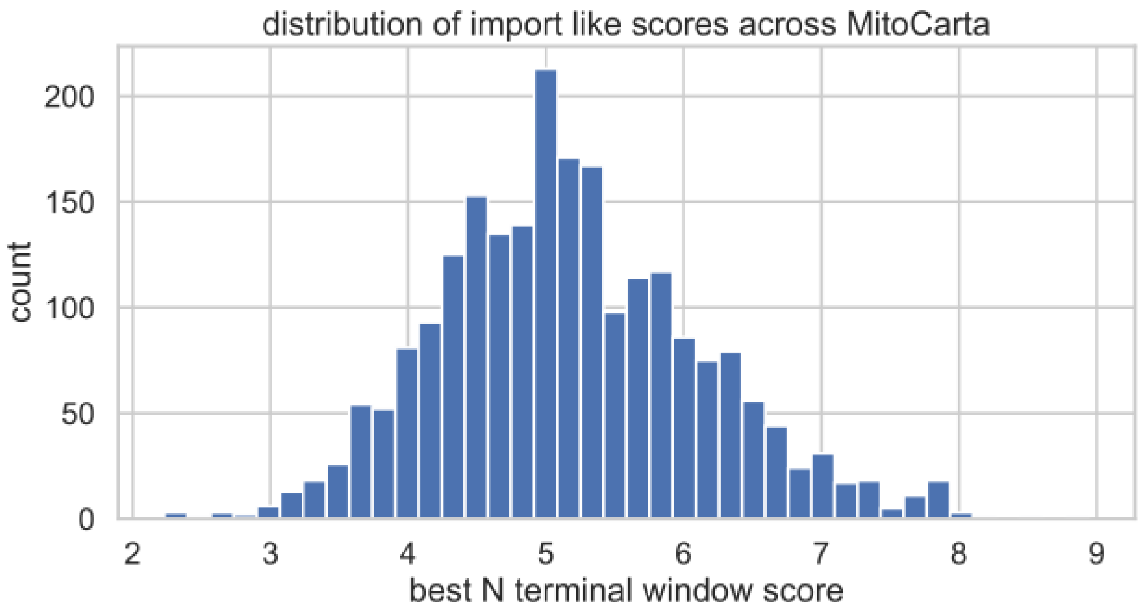


Figure 4. Minimal-edit mutagenesis tunes mitochondrial import signals. Profiles show hydropathy, charge, and helix propensity for N-terminal sequence windows before and after small amino acid edits, demonstrating that mitochondrial presequences are tunable through minimal sequence changes. (A-B) NP_659418: Edits increase charge and helix propensity, strengthening the predicted import signal. (C-D) NP_060599: Edits enhance amphipathic helix balance, raising import potential. (E-F) NP_976043: Edits adjust hydropathy-charge balance, improving predicted targeting efficiency.

A



B

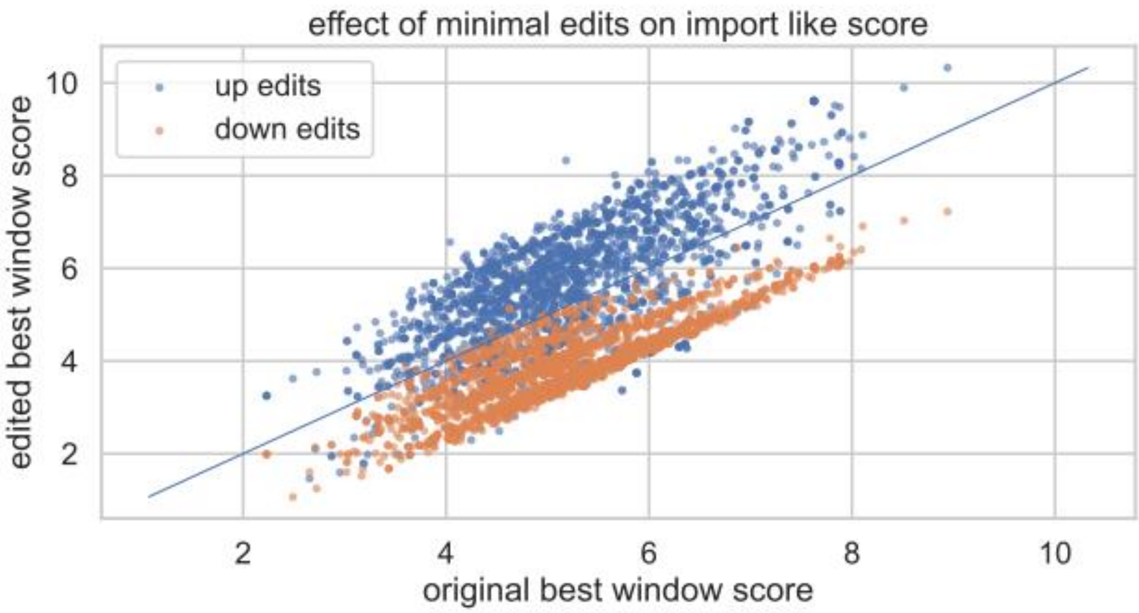


Figure 5. Global distribution of import-like scores. (A) Histogram of predicted import scores across MitoCarta proteins reveals a unimodal distribution, with most proteins scoring between 4-7 and a minority exceeding 8. (B) Scatterplot of edit effects demonstrates that small amino acid changes can substantially increase or decrease import predictions.

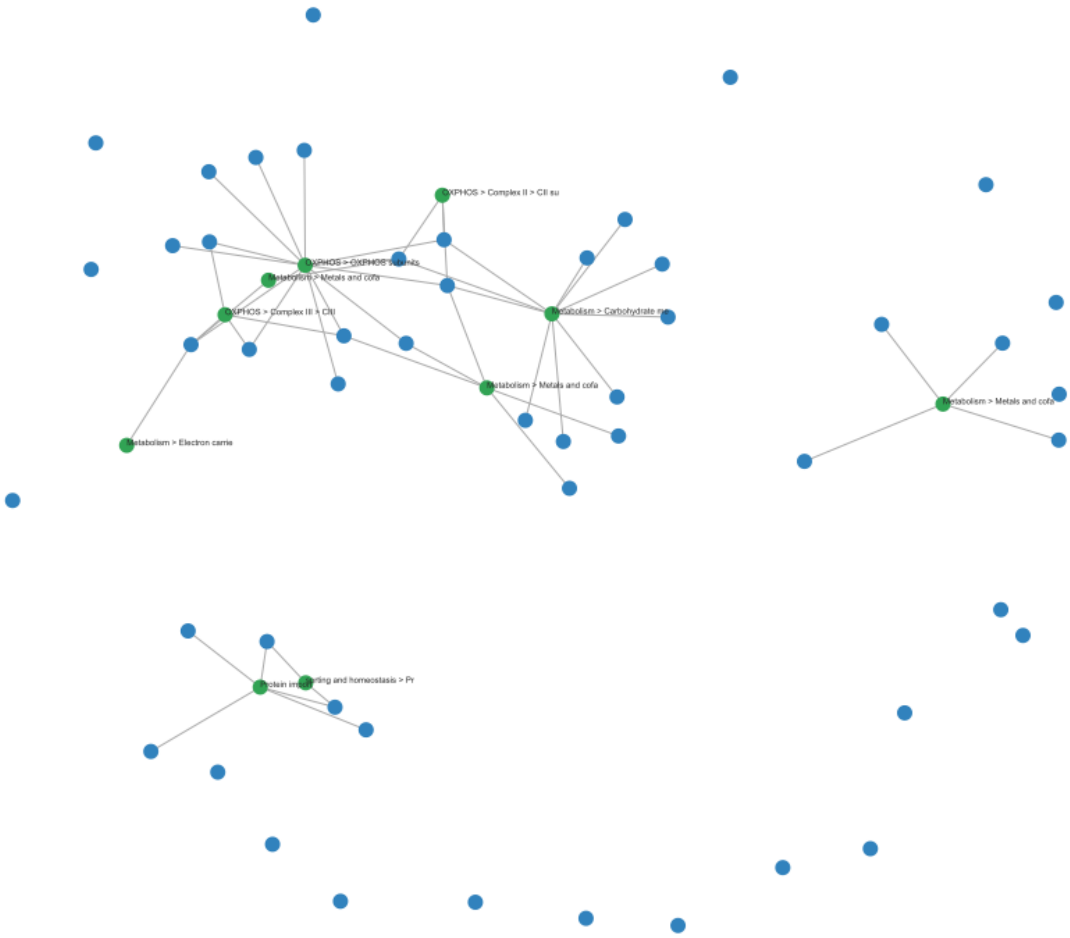


Figure 6. Network organization of mitochondrial pathways. Bipartite network of proteins (nodes) and pathways shows highly connected hubs. Oxidative phosphorylation, translation, and import dominate. Graph visualization highlights dense connectivity of hub pathways compared to smaller peripheral modules.

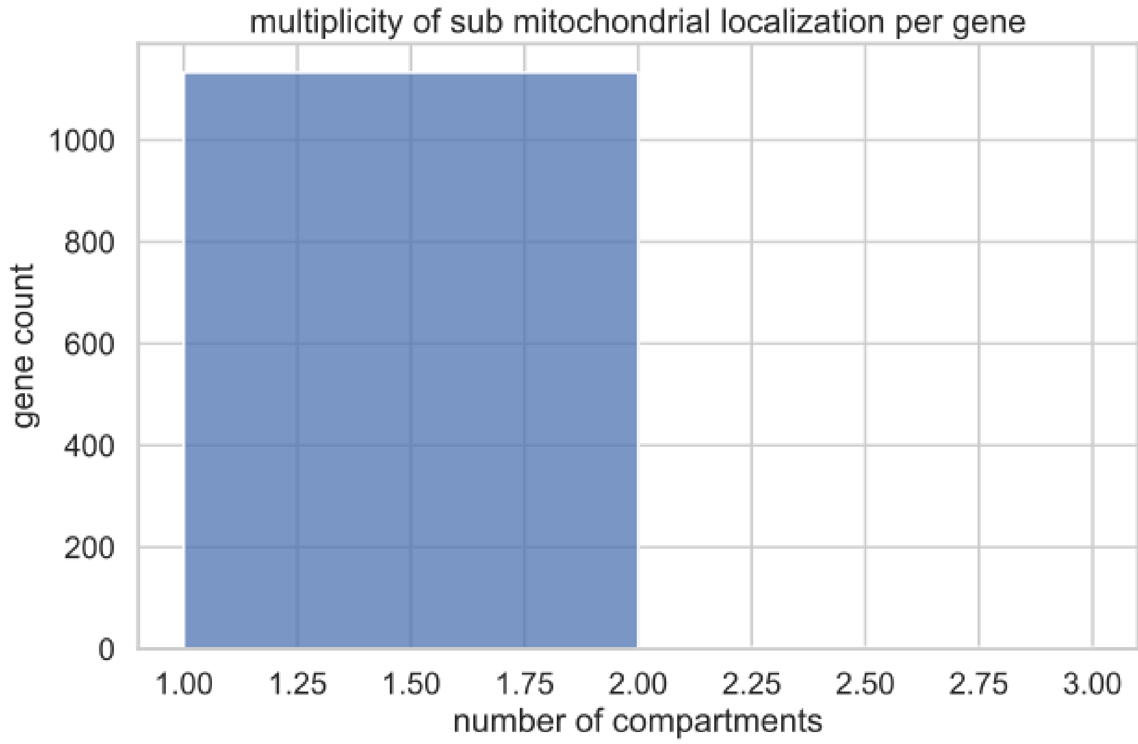


Figure 7. Dual-localization proteins reveal multifunctionality. Histogram shows ~80% of proteins localized to one compartment, ~15% to two compartments, and ~5% to three or more.